

Deep Learning

Data Science

Course Schedule

1. Exploratory Data Analysis
2. Clustering
3. Probabilistic Models
4. Statistical Inference
5. Machine Learning
6. Feature Engineering
7. Model Evaluation
8. Deep Learning
9. Time Series
10. Causality

Deep Learning

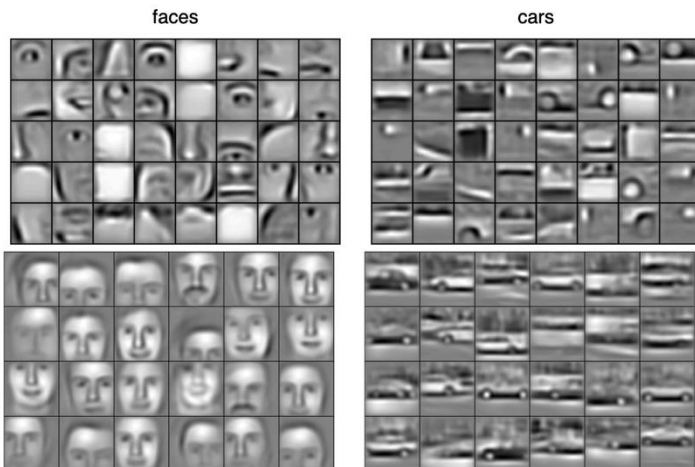
- neural networks
- network training
- convolutional neural networks
- recurrent neural networks
- tabular vs unstructured data

Ladder of Generalization

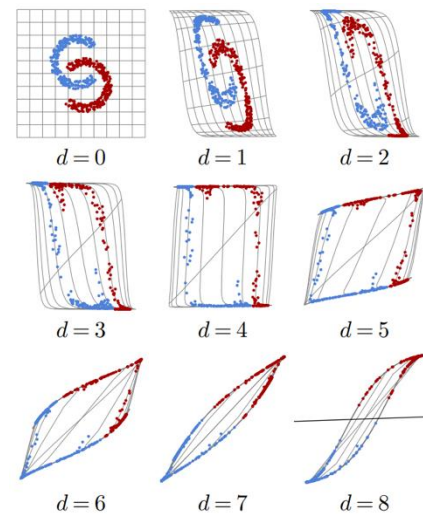
classic ML: feature engineering

deep learning: feature learning

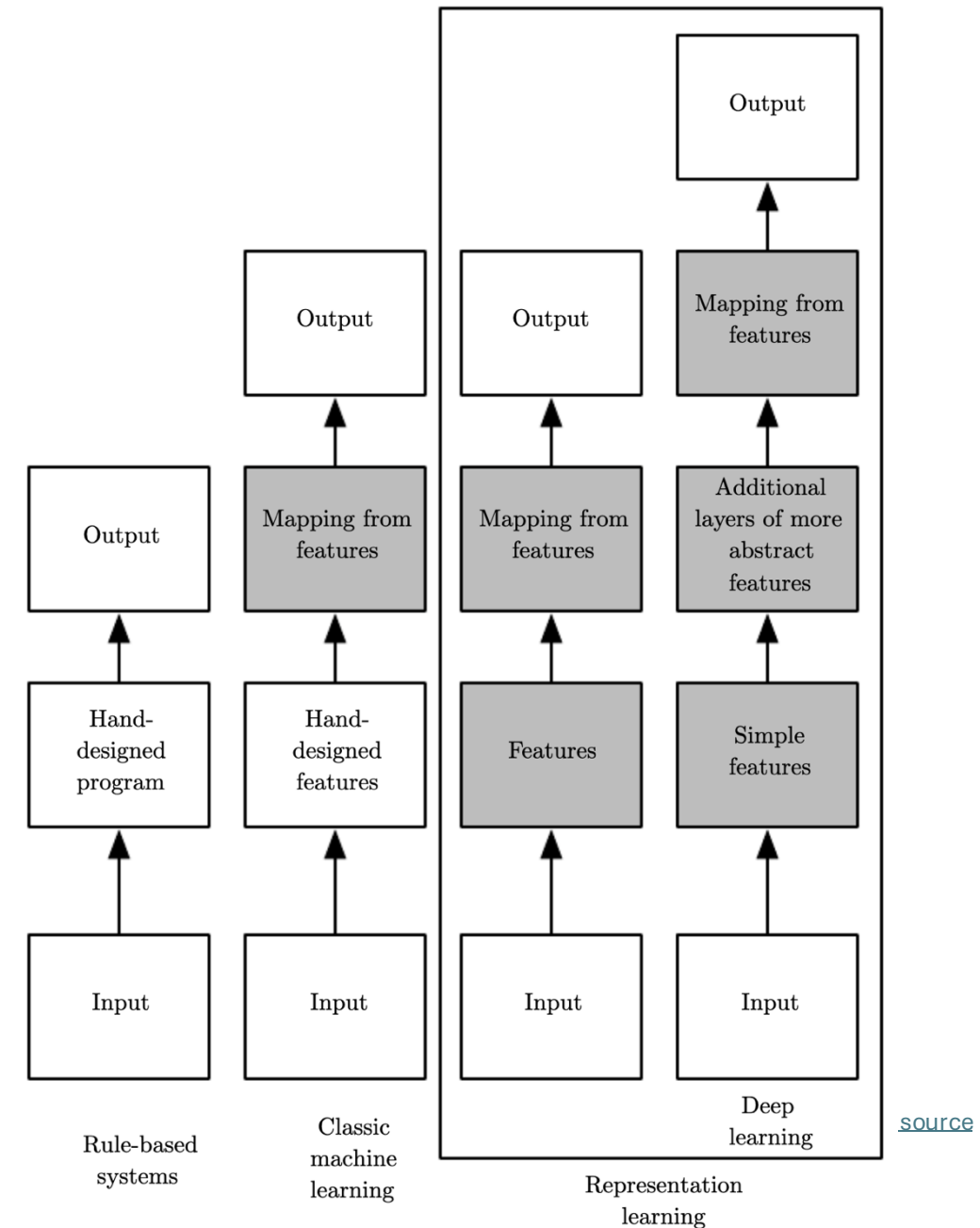
(hierarchy of concepts learned from raw data in deep graph with many layers)



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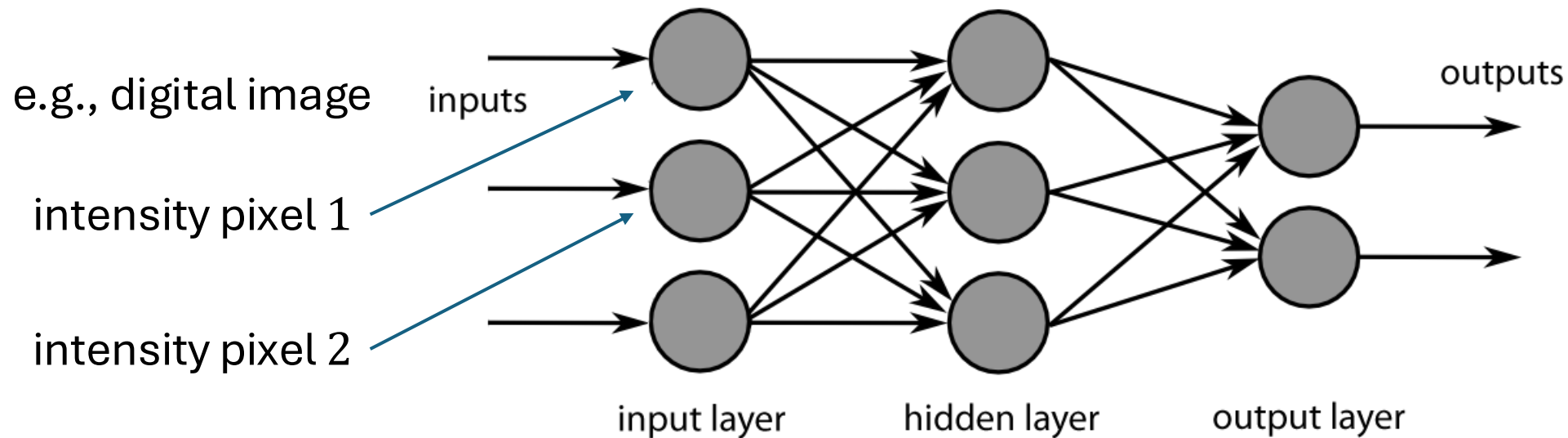
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Neural Networks

idea: powerful algorithm by combining many linear building blocks



each node exchanges information only with directly connected nodes
→ parallel computation

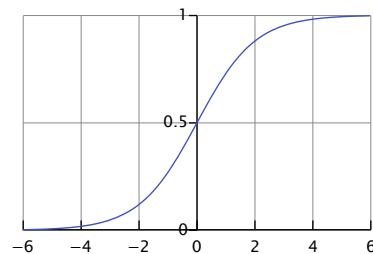
Artificial Neuron

linear model with parameters called weights $\mathbf{w}$ (including bias, not shown for simplicity)

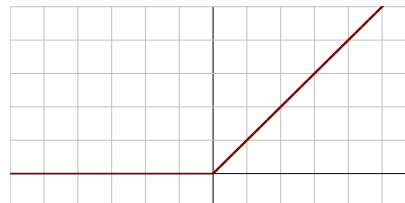
non-linear via (differentiable) activation function on top of linear model

examples:

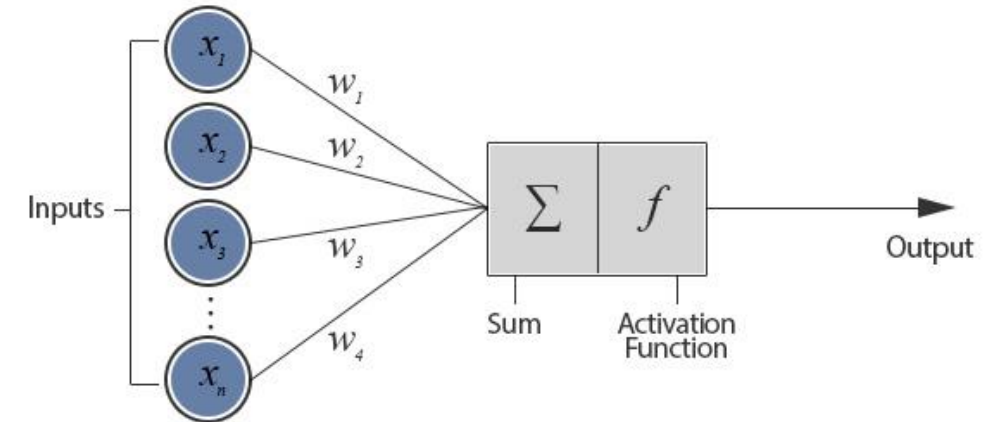
- sigmoid



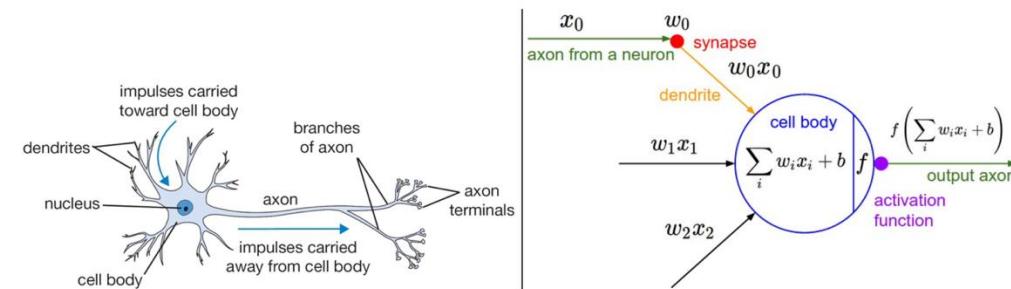
- ReLU (Rectified Linear Unit)



artificial neuron (node in neural network):

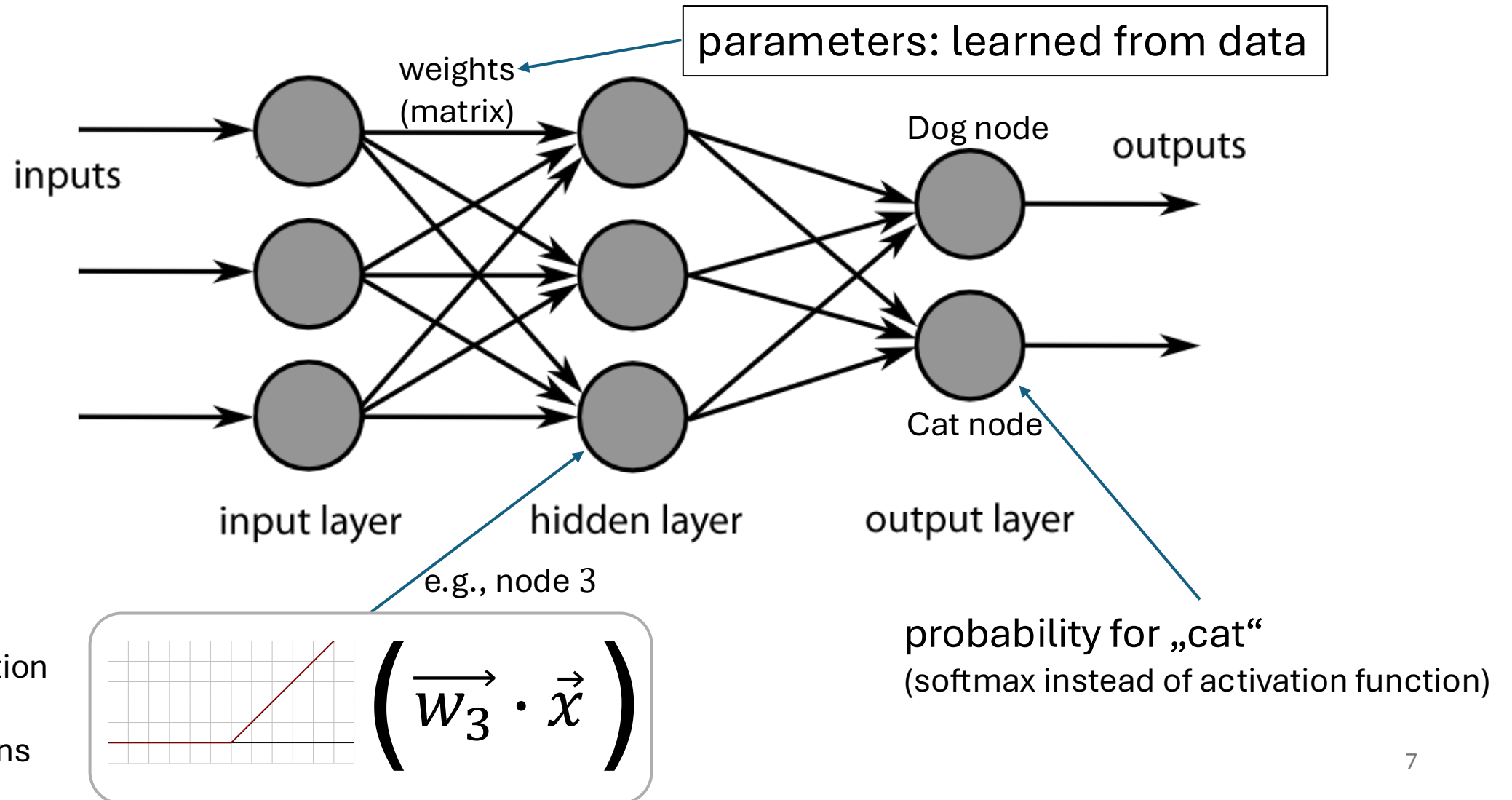


inspired from biological neurons:



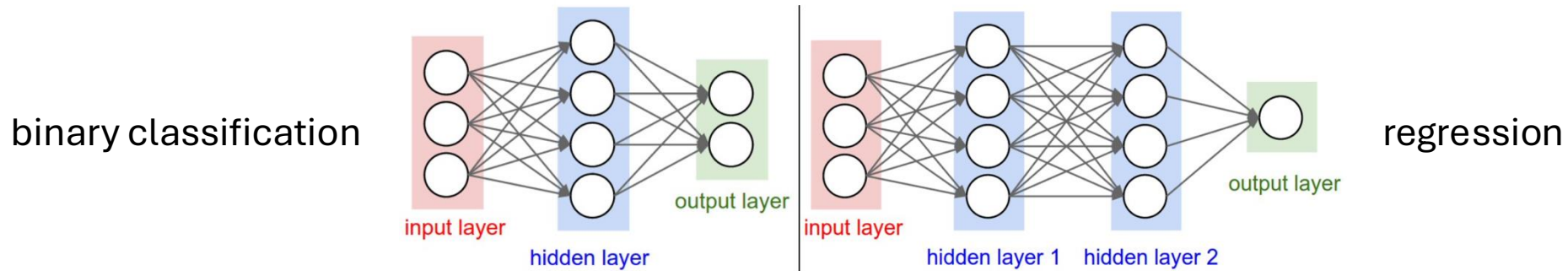
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Neural Network as Nonlinear Function



Multi-Layer Perceptron (MLP)

fully-connected feed-forward network with at least one hidden layer
(universal function approximator)



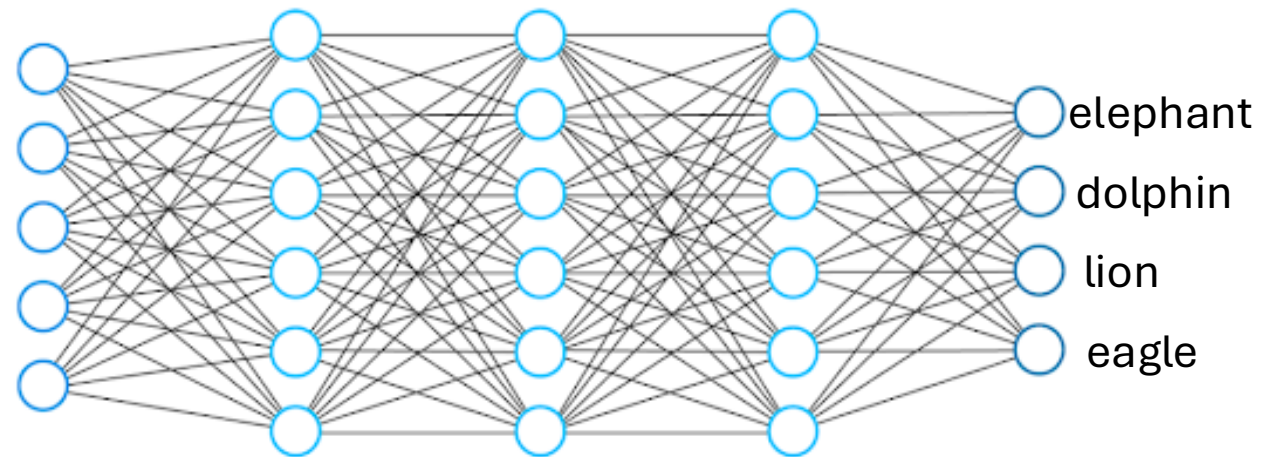
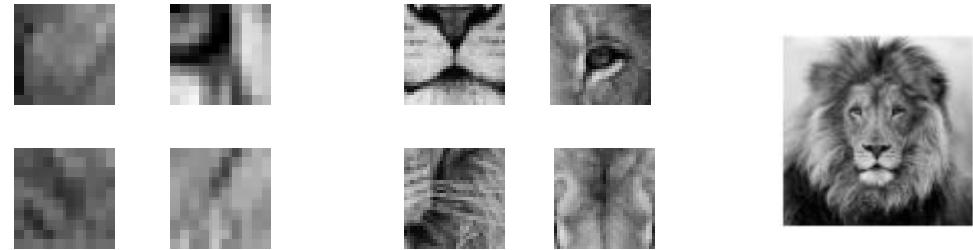
toward deep learning: add hidden layers

more layers (depth) more efficient than just more nodes (width):
less parameters needed for same function complexity

Hierarchical Representation

deep learning:
several hidden layers

increasing abstraction
→



Classification Networks

cross-entropy loss (one output node for each class k):

$$L(y, \hat{y}) = - \sum_{k=1}^K y_k \log \hat{y}_k$$

using softmax on output nodes:

$$\hat{y}_k = \frac{e^{o_k}}{\sum_{l=1}^K e^{o_l}}$$

→ prediction of probabilities

regression networks:

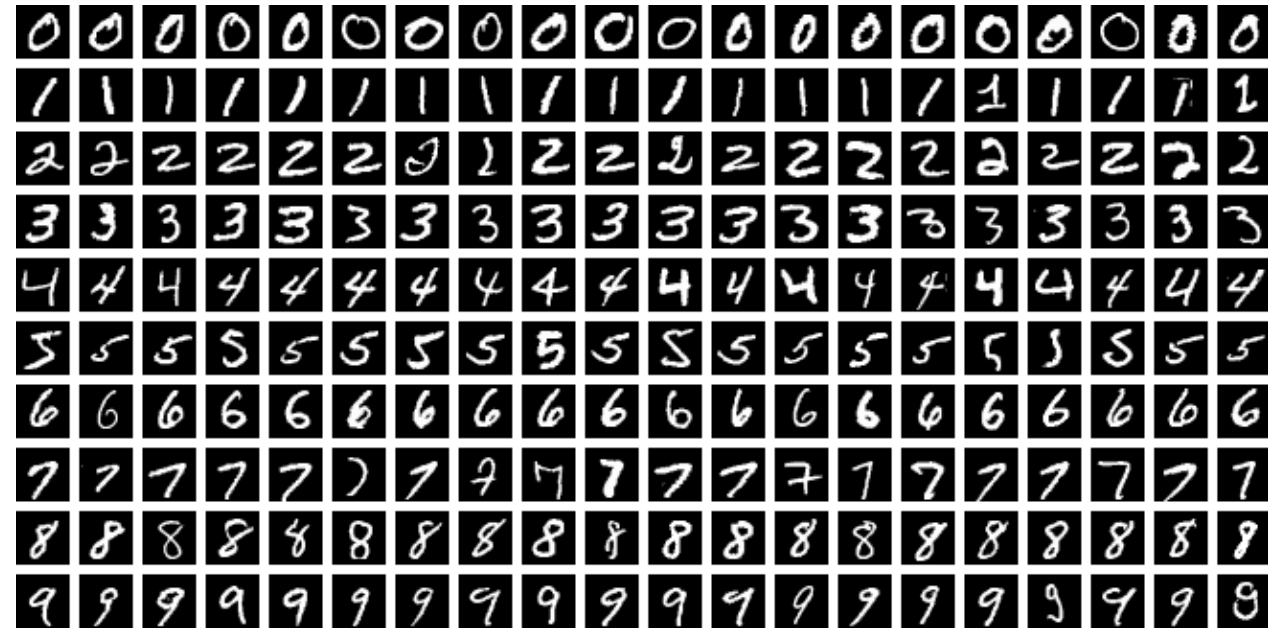
- squared error loss:
 $L(y, \hat{y}) = (y - \hat{y})^2$
- just one output node
- no function on output
→ prediction of real numbers

Image Classification: Lots of Categories



MNIST

- Modified National Institute of Standards and Technology
- handwritten digits (10 classes)
- black and white images
- 28×28 pixels
- 60k training and 10k test images



CIFAR-10

- Canadian Institute for Advanced Research
- 10 different labeled object classes
- color images
- 32×32 pixels
- 50k training and 10k test images

airplane



automobile



bird



cat



deer



dog



frog



horse



ship

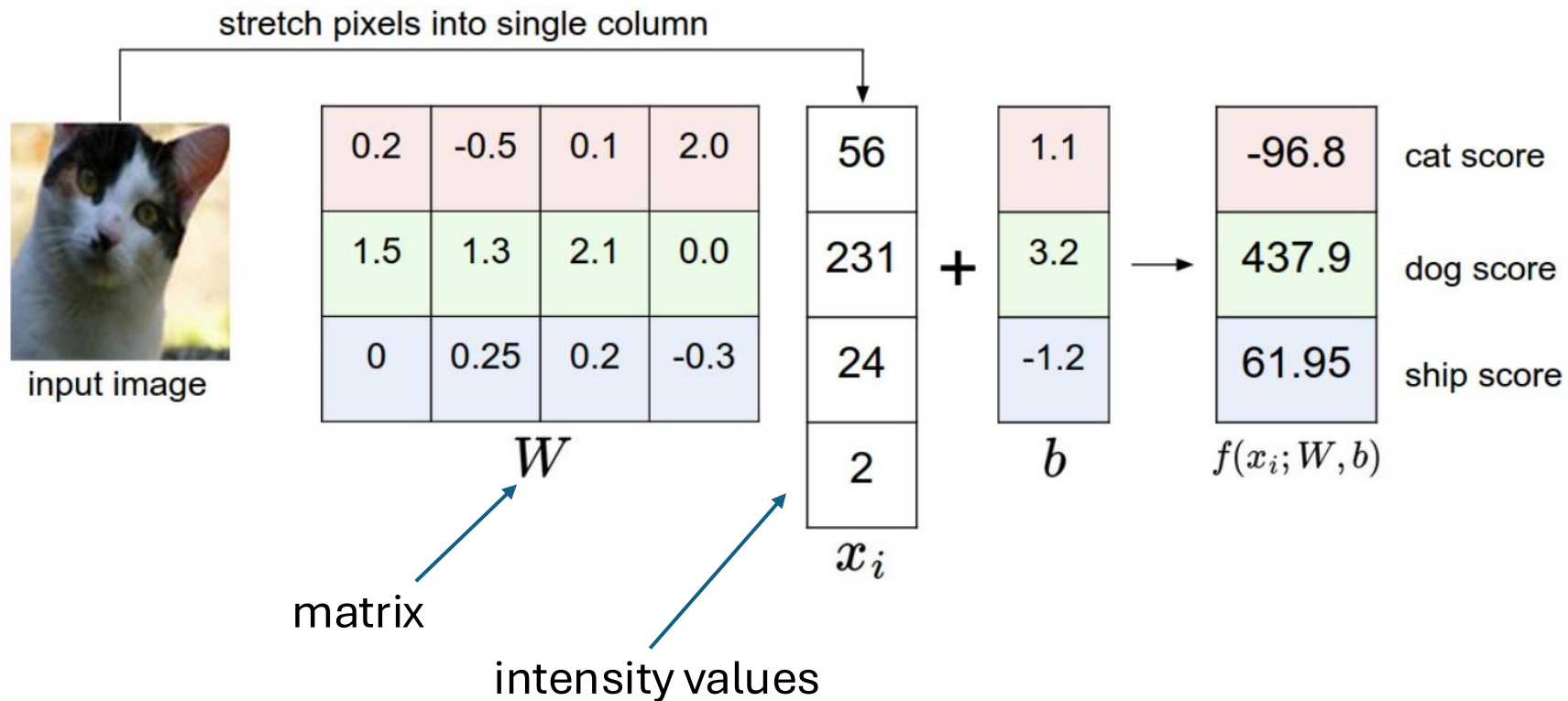


truck



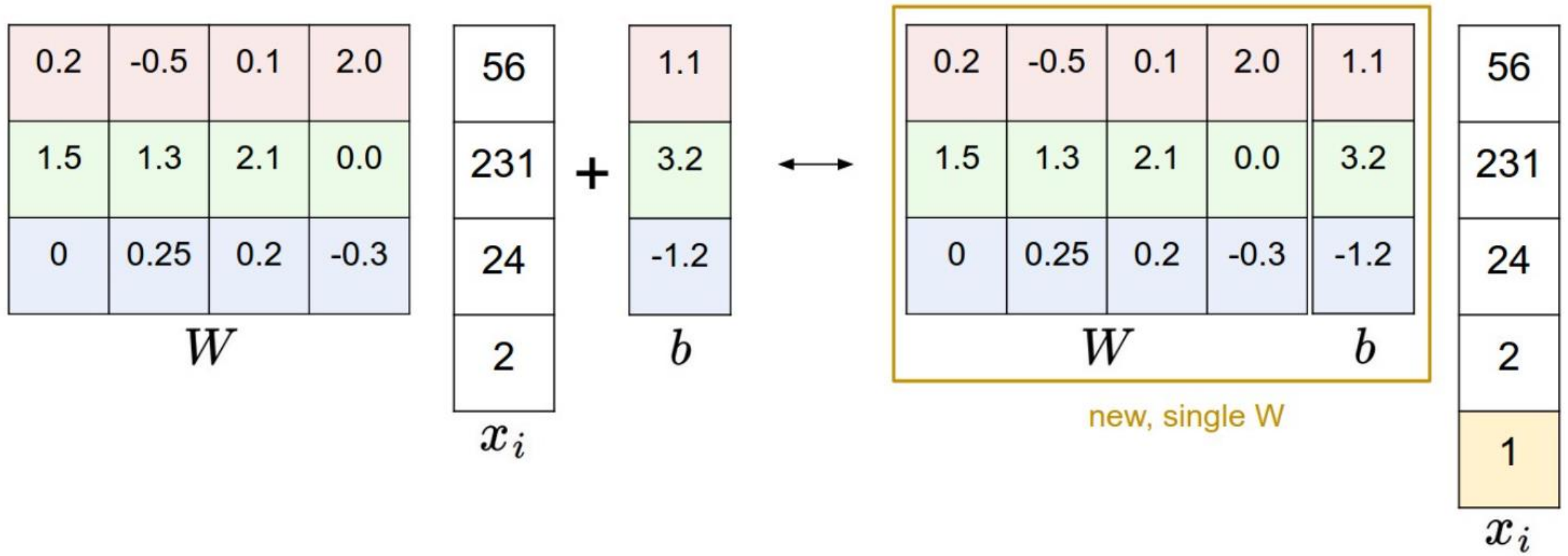
Image Classification with Linear Regression

simplified example: 4-pixel image, 3 classes



need for one common image size per model

Simplified Matrix Multiplication: Bias Trick



geometric interpretation:
separating hyperplanes



matching interpretation:
generic class templates

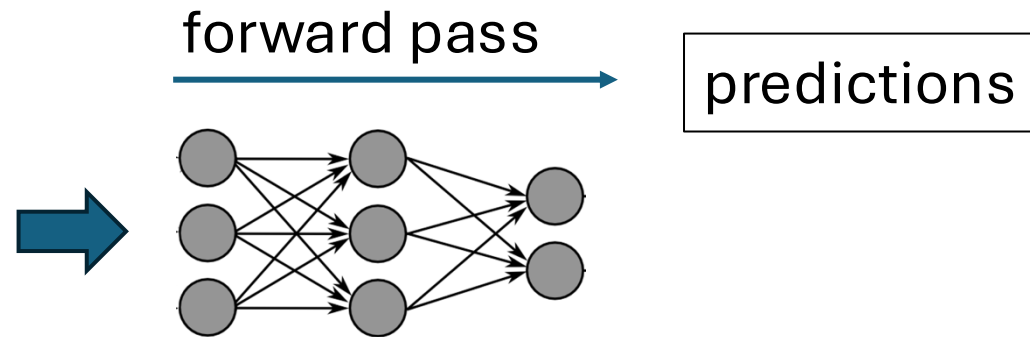
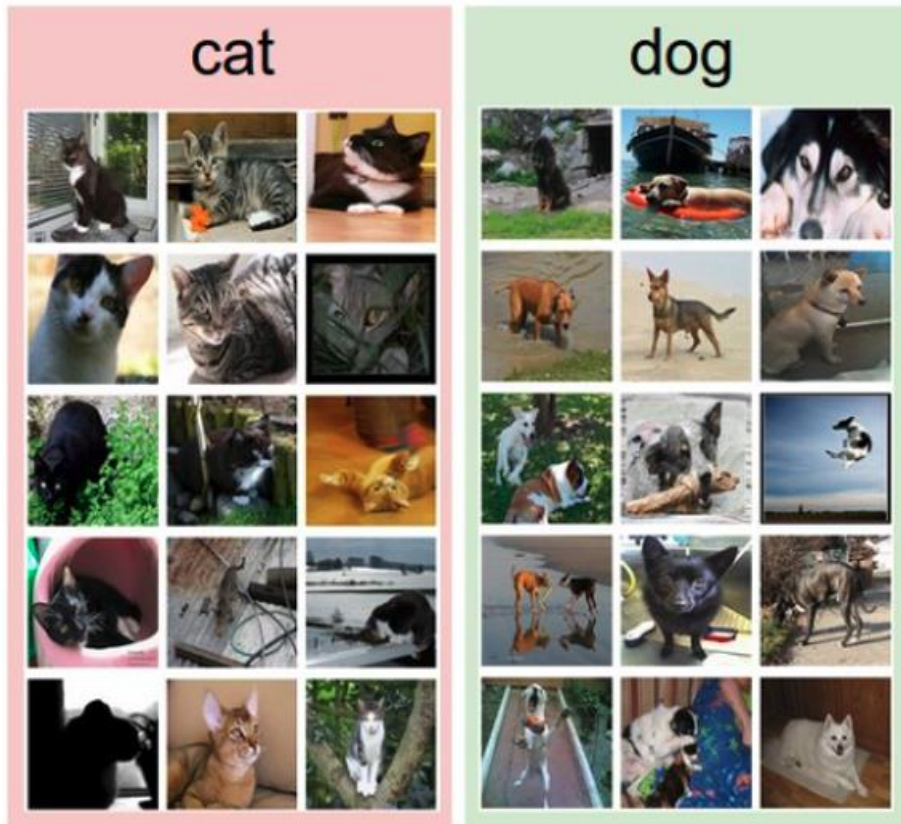


different rows in matrix W

raw-pixel images not linearly separable
→ linear model has not enough representational power

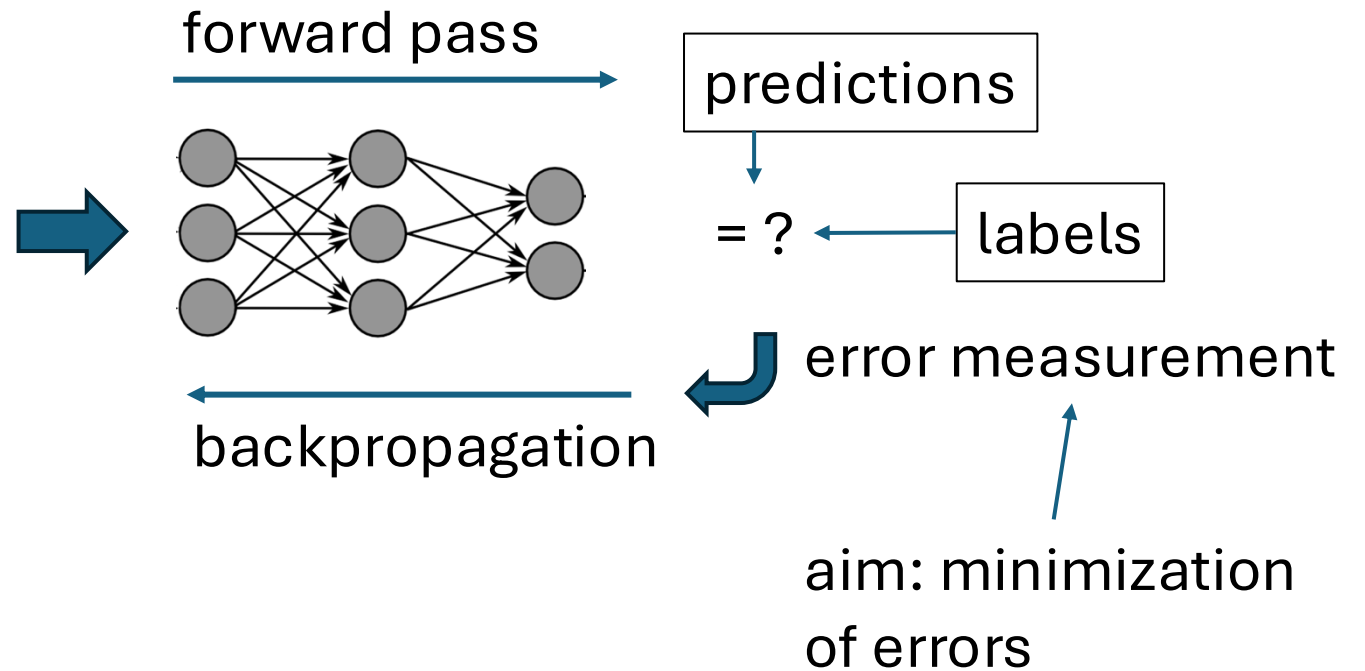
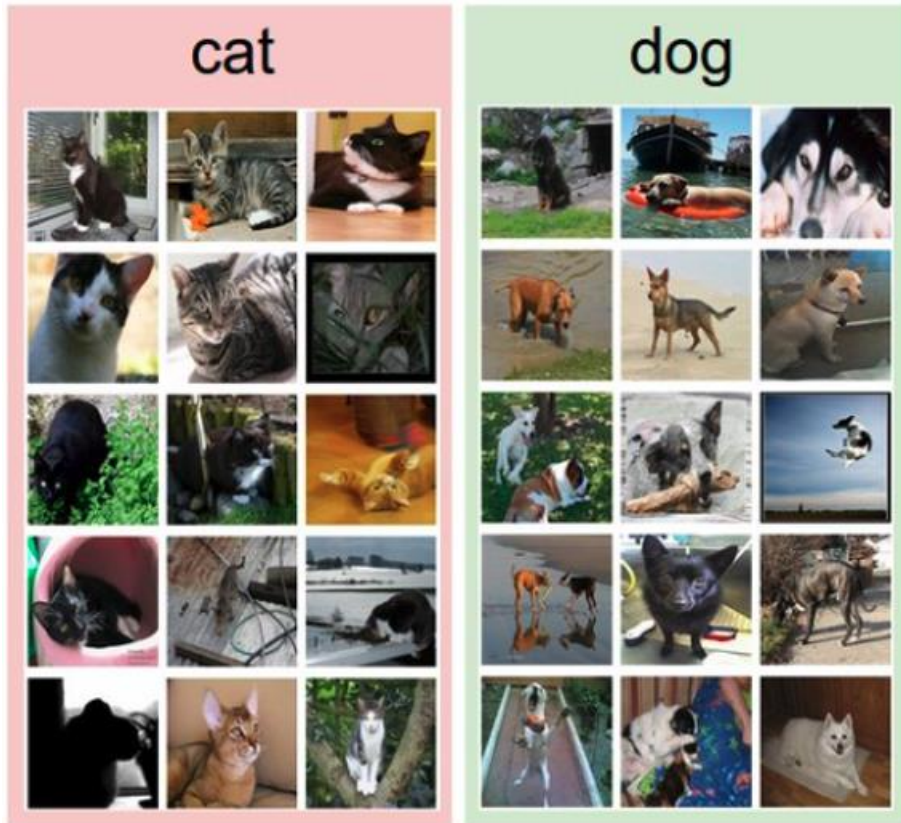
Training with Labeled Images

label:



Training with Labeled Images

label:



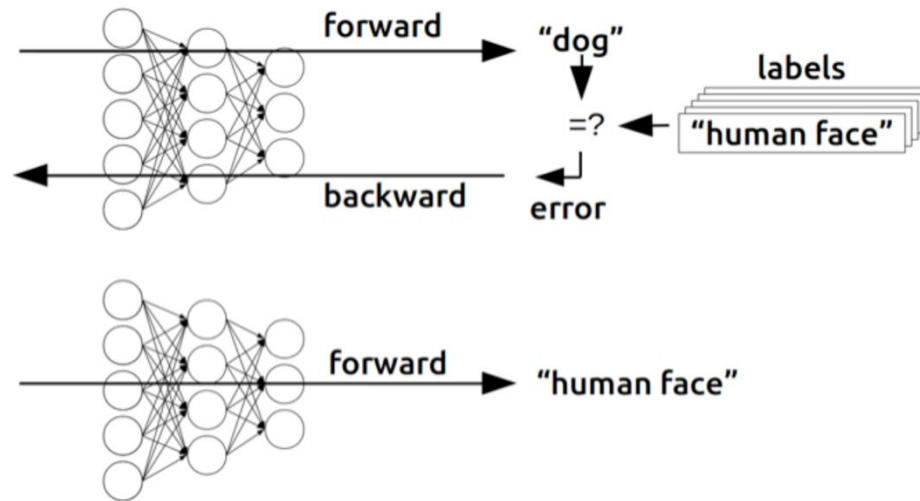
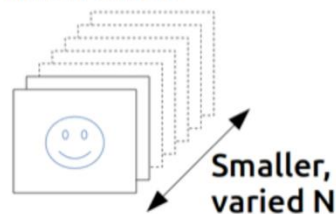
Find Gradients for (Deep) Neural Networks

backpropagation of errors through network layers via chain rule of calculus (enables learning of deep neural networks)

Training



Inference



[source](#)

forward pass:

- fix current weights
- compute predictions

backward pass:

- compute errors
- calculate gradients (backpropagation of errors)
- update weights accordingly (gradient descent)

Training: Iterative Adjustment of Weights

```
for epoch in range(epochs):  
    for X, y in mini_batches:  
        pred = model(X)  
        cost = loss(pred, y)  
        cost.backward()  
        optimizer.step()
```

quantification of error
(here: cross entropy)

chain rule to calculate
gradient (direction) of
cost function with
respect to weights

one step in direction of steepest
descent: (stochastic) gradient descent

Example WOLOG

- regression (squared error loss, identity output function g)
- with one hidden layer ($\mathbf{w}$: α, β)

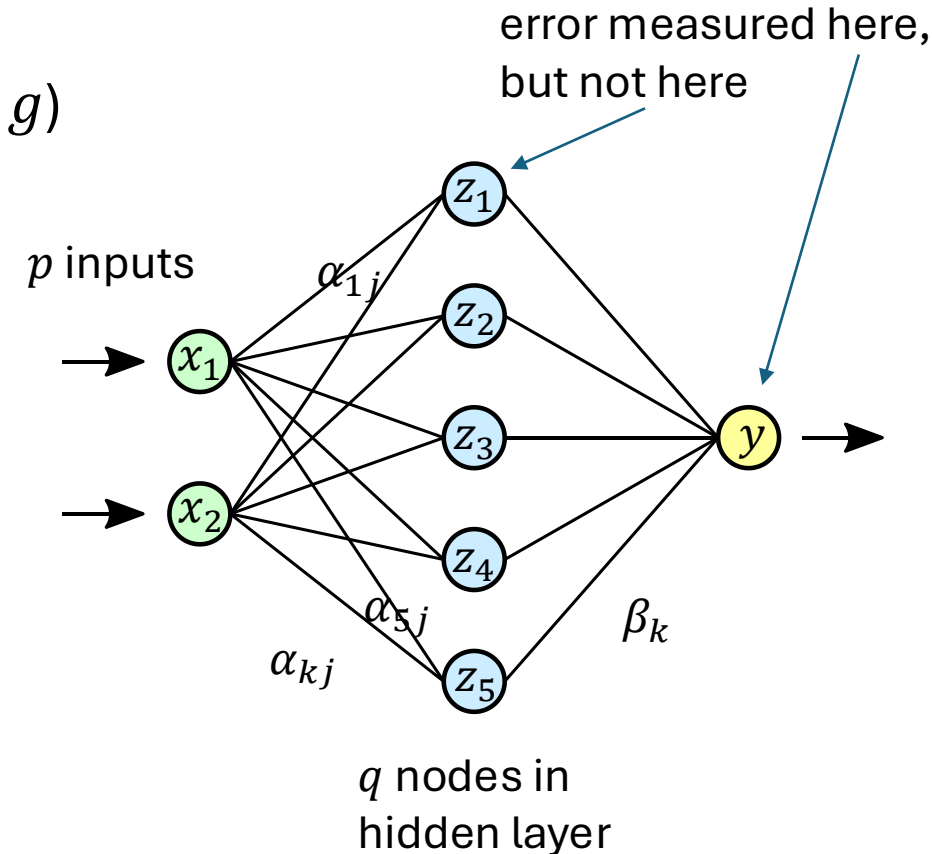
$$\hat{y}_i = f(\mathbf{x}_i; \mathbf{w}) = g(\mathbf{z}_i; \beta) = \sum_{k=0}^q \beta_k z_{ik}$$

$$z_{ik} = h(\mathbf{x}_i; \alpha_k) = h\left(\sum_{j=0}^p \alpha_{kj} x_{ij}\right)$$

activation
function

cost function:

$$\mathcal{L}(\mathbf{w}) = \sum_{i=1}^n L_i(y_i, \hat{y}_i; \mathbf{w}) = \sum_{i=1}^n (y_i - f(\mathbf{x}_i; \alpha, \beta))^2$$



Example WOLOG

gradients:

$$\frac{\partial L_i}{\partial \beta_k} = -2 (y_i - \hat{y}_i) z_{ik} = \delta_i z_{ik}$$
$$\frac{\partial L_i}{\partial \alpha_{kj}} = -2 (y_i - \hat{y}_i) \beta_k h'_k \left(\sum_{j=0}^p \alpha_{kj} x_{ij} \right) x_{ij} = s_{ik} x_{ij}$$

backpropagation equations: $s_{ik} = h'_k \left(\sum_{j=0}^p \alpha_{kj} x_{ij} \right) \beta_k \delta_i$

use errors of later layers to calculate errors of earlier ones (avoiding redundant calculations of intermediate terms)

Using Gradients for Iterative Learning

use gradients found via backpropagation to iteratively update weights (gradient descent):

$$\beta_k^{(r+1)} = \beta_k^{(r)} - \eta_r \sum_{i=1}^n \frac{\partial L_i}{\partial \beta_k^{(r)}}$$

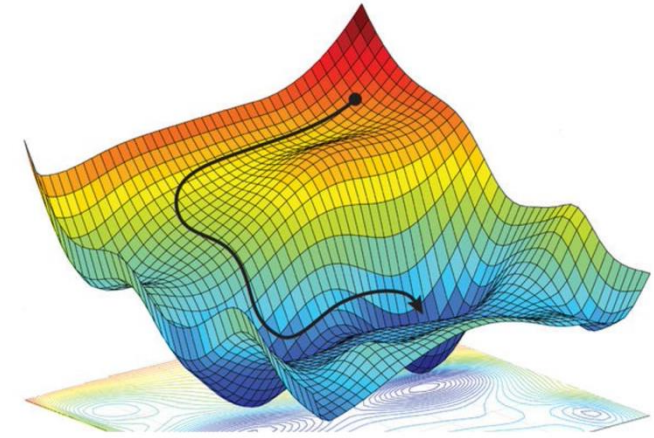
$$\alpha_{kj}^{(r+1)} = \alpha_{kj}^{(r)} - \eta_r \sum_{i=1}^n \frac{\partial L_i}{\partial \alpha_{kj}^{(r)}}$$

- adaptive learning rate η_r : learning rate often adjusted per iteration
- weight initialization: choose small random weights as starting values to break symmetry (typically using some heuristics)

Stochastic Gradient Descent (SGD)

weight updates: $\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla_{\mathbf{w}} \mathcal{L}$

step size η cost function $\mathcal{L}$



options: all training examples

- $\mathcal{L} = \frac{1}{n} \sum_{i=1}^n L_i$

batch gradient descent (whole training dataset)

- $\mathcal{L} = L_i$

stochastic gradient descent (single example)

- $\mathcal{L} = \frac{1}{m} \sum_{i=1}^m L_i$

mini-batch stochastic gradient descent

mini-batch size $m \rightarrow$ hyperparameter: tradeoffs between runtime & memory, convergence & generalization

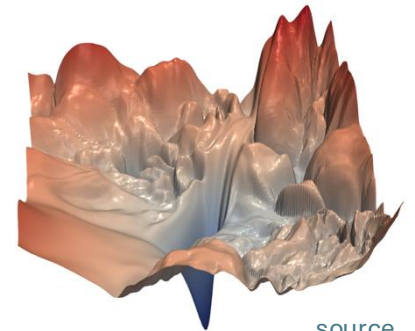
random fluctuations of SGD help to escape saddle points

How to Train Deep Neural Networks Effectively?

optimization and regularization difficult

- local vs global minima, saddle points, easily overfitting
- many hyperparameters to tune

typical loss surface:



[source](#)

but many methods to get it working in practice (despite partly patchy theoretical understanding)

- optimization: activation and loss functions, weight initialization, adaptive learning rate (e.g., Adam), learning rate schedulers
- explicit regularization: weight decay, dropout, data augmentation
- implicit regularization: early stopping, batch/layer normalization, SGD

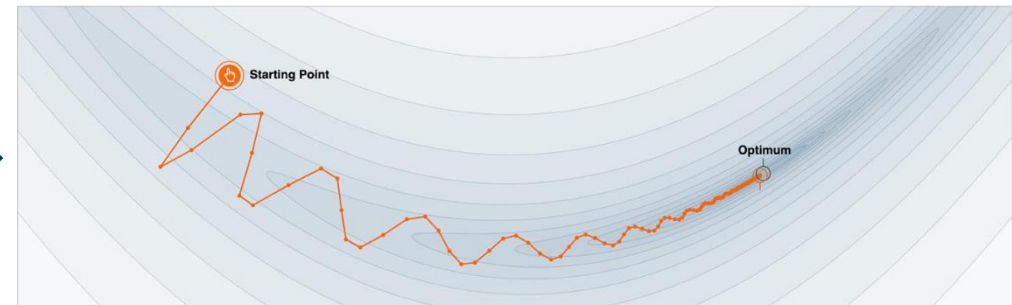
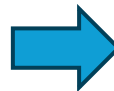
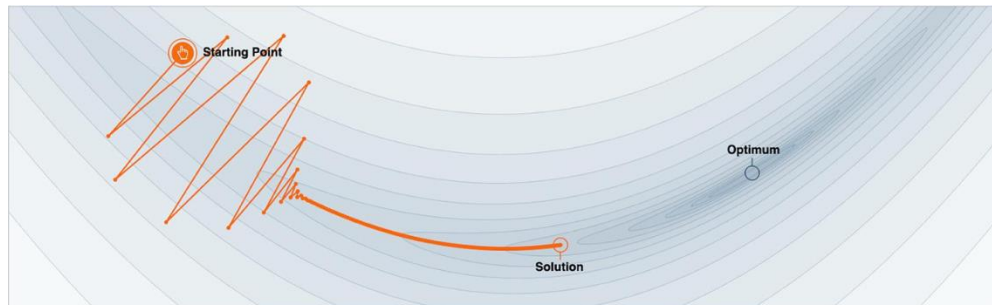
Gradient Descent with Momentum

algorithm:

- estimate gradient: $\mathbf{g} = \nabla_{\mathbf{w}} \mathcal{L}$
- compute velocity update: $\mathbf{v} \leftarrow \alpha \mathbf{v} - \eta \mathbf{g}$
- apply weight update: $\mathbf{w} \leftarrow \mathbf{w} + \mathbf{v}$

hyperparameter ($0 < \alpha < 1$)
specifying exponential decay

→ no bouncing around of weights, escape from local minima and saddle points



Adaptive Learning Rate

better convergence by adapting learning rate for each weight per iteration:
lower/higher learning rates for weights with large/small updates

popular methods (g_w denotes component of gradient for **individual weight** w):

- Adagrad: $w \leftarrow w - \frac{\eta}{\sqrt{\sum_{\tau=1}^t g_{w,\tau}^2}} g_w$ with t, τ denoting current and past iterations
(issue: sum in denominator grows with more iterations $\rightarrow$ can get stuck)
- RMSProp: $w \leftarrow w - \frac{\eta}{\sqrt{v(w)}} g_w$ with $v(w) \leftarrow \gamma v(w) + (1 - \gamma) g_w^2$
- Adam (Adaptive Moment Optimization): combines RMSProp with momentum

Learning Rate Schedulers

learning rate η in gradient descent typically set to small value (e.g., 0.001)

use momentum and adaptive learning rates to improve optimization

learning rate scheduling as further alternative (can be used on top):

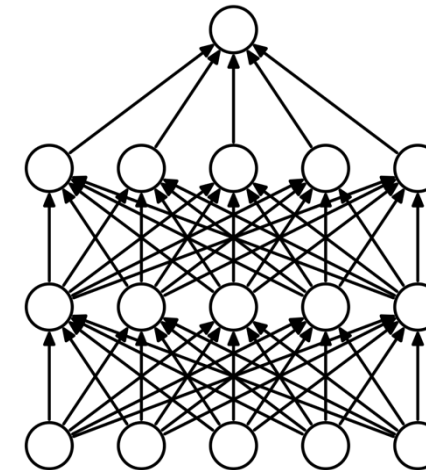
- decay by a certain factor every given number of epochs
- reduce when a metric has stopped improving
- smooth decay by cosine annealing (optionally cyclic with restarts)
- ...

Dropout in Neural Networks

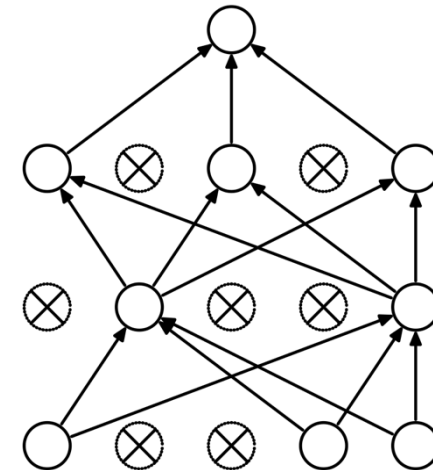
goal: prevent overfitting of large neural networks

randomly drop non-output nodes (along with their connections) during training (not prediction)
→ adaptability: regularizing each hidden node to perform well regardless of which other hidden nodes are in the model

- for each mini-batch, randomly sample independent binary masks for the nodes
- destroying extracted features rather than input values



(a) Standard Neural Net



(b) After applying dropout.

[source](#)

Batch Normalization

Input: Values of x over a mini-batch: $\mathcal{B} = \{x_{1\dots m}\}$;

Parameters to be learned: γ, β

Output: $\{y_i = \text{BN}_{\gamma, \beta}(x_i)\}$

$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m} \sum_{i=1}^m x_i \quad // \text{ mini-batch mean}$$

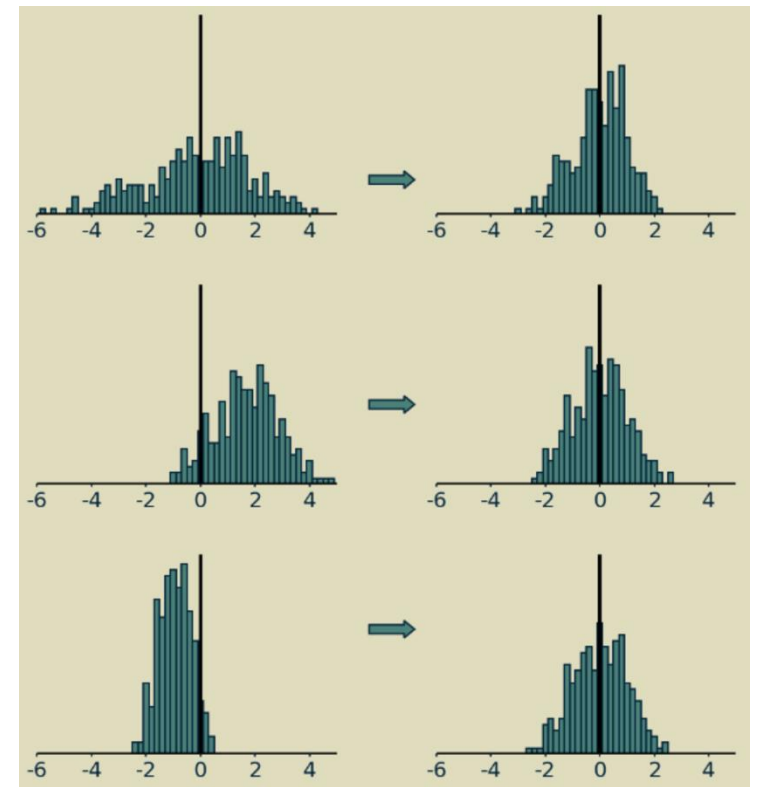
$$\sigma_{\mathcal{B}}^2 \leftarrow \frac{1}{m} \sum_{i=1}^m (x_i - \mu_{\mathcal{B}})^2 \quad // \text{ mini-batch variance}$$

$$\hat{x}_i \leftarrow \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \quad // \text{ normalize}$$

$$y_i \leftarrow \gamma \hat{x}_i + \beta \equiv \text{BN}_{\gamma, \beta}(x_i) \quad // \text{ scale and shift}$$

Algorithm 1: Batch Normalizing Transform, applied to activation x over a mini-batch.

[source](#)



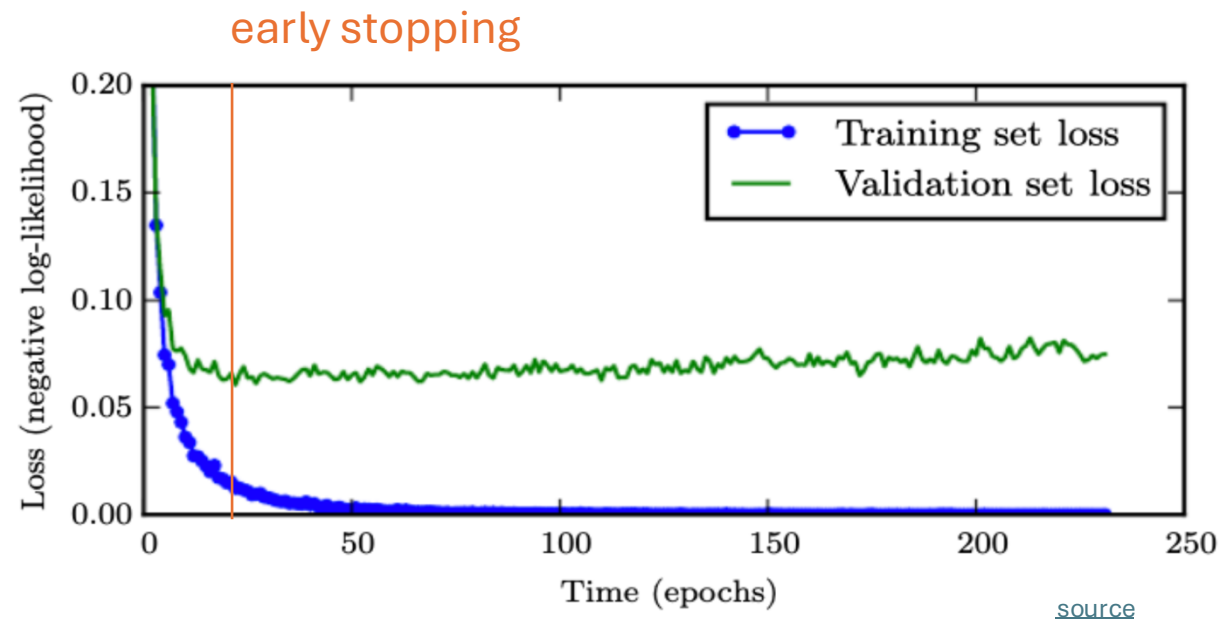
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adaptive reparameterization of inputs to network layer (before or after activation)
independently for each feature

(implicit) regularization effect

Early Stopping

loss independently measured on validation set
halting training when overfitting begins to occur



From i.i.d. Assumptions to Structured Deep Learning

fully-connected networks treat inputs as independent features (i.i.d.): no built-in notion of structure or ordering (example: pixels in an image treated as unrelated numbers)

but real data is often structured → strong correlations between samples

- images: nearby pixels are highly correlated
- text / speech: words and sounds depend on previous context
- time series: observations depend on earlier time steps

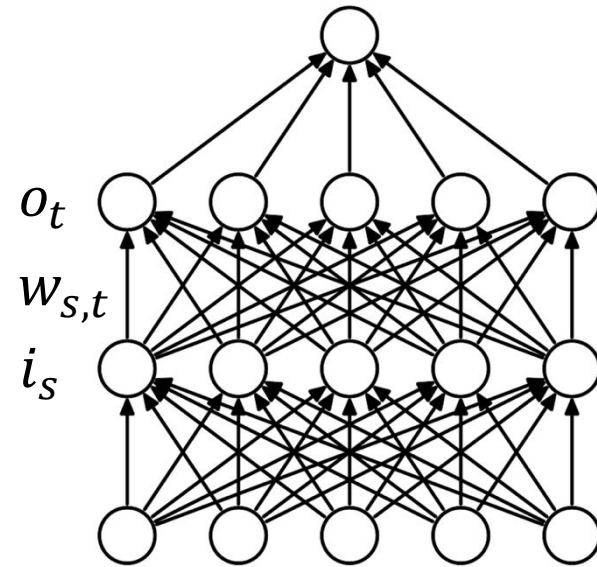
ignoring these correlations leads to suboptimal models → CNNs, RNNs

Structure of Fully-Connected Neural Networks

computation in fully-connected network:
matrix multiplication of scalar inputs and weights

$$o_t = \sum_s w_{s,t} i_s$$

dropped dimension of different training observations
(in mini-batch) in this view



Convolutional Neural Networks (CNN or ConvNet)

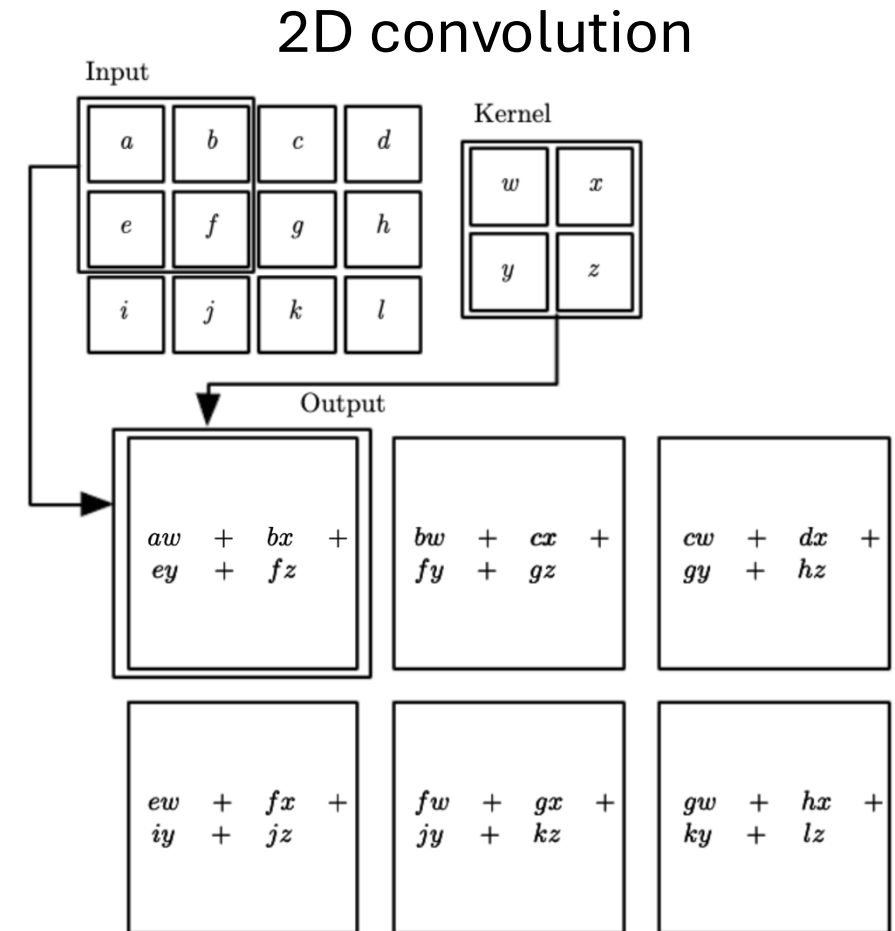
actually, correlation (convolution would have $-$ here):

$$o_{x,y} = \sum_{s,t} w_{s,t} i_{x+s,y+t}$$

feature map (transformed image) kernel (learned) image (input or feature map)

(again, dropping dimension of different training observations)

→ exploit local spatial correlations



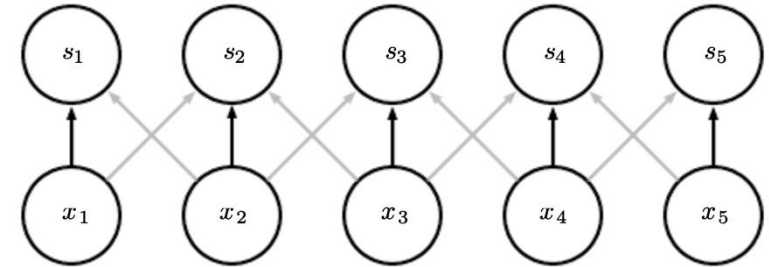
Regularization Effects

- sparse interactions: much less weights
- parameter sharing: use same weights for different connections (locations)

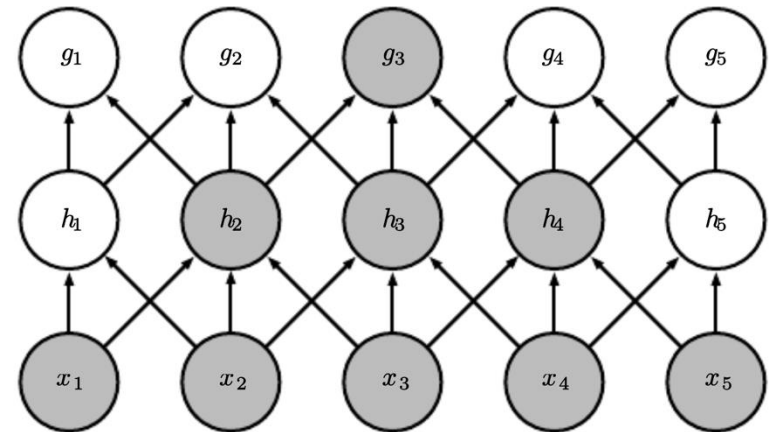
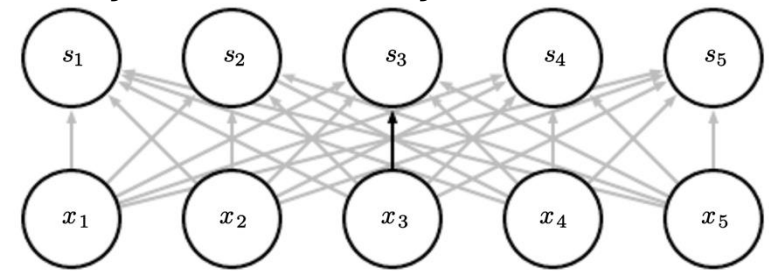
effect of receptive field over several layers:

- consider only locally restricted number of input values from previous layer
 - grows for earlier layers (indirect interactions)
- hierarchical patterns from simple building blocks (many aspects of nature hierarchical)

convolutional layer:



fully-connected layer:

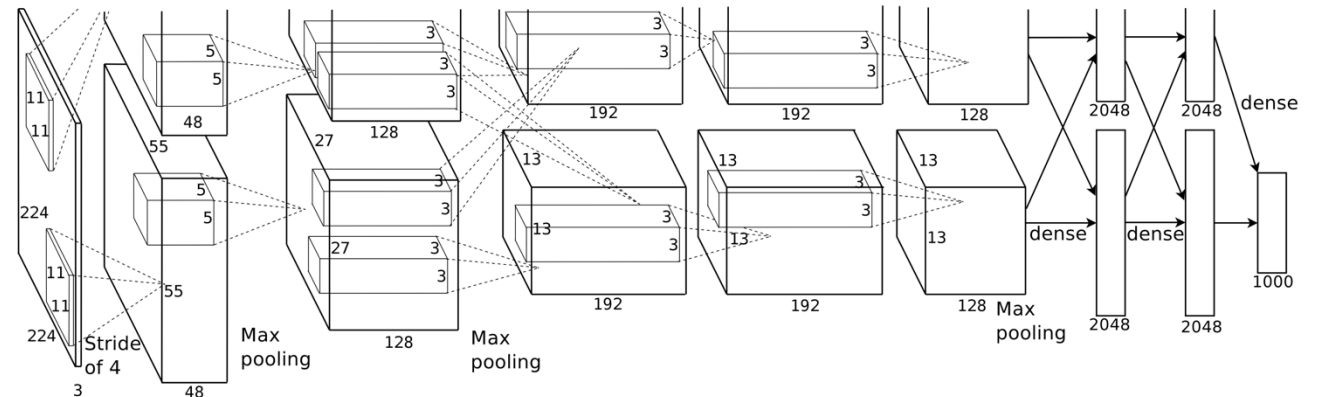
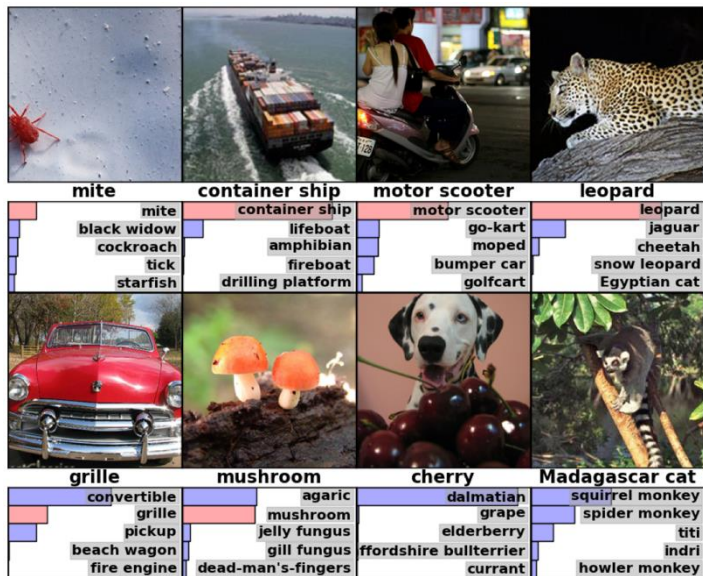


Rise of Deep Learning

a little bit oversimplified:

deep learning = lots of training data + parallel computation + smart algorithms

AlexNet: ImageNet (with data augmentation) + GPUs + ReLU, dropout, SGD



[source](#)

Transfer Learning

problem with ConvNets trained from scratch (in fact, with all ML methods): only suited for domain of training data

idea of transfer learning:

pretrain a big model (called foundation model) on a broad data set, and then use these learnings for subsequent trainings on specific (typically, narrow) data

two forms of transfer learning: feature extraction and finetuning

Feature Extraction

approach:

- get a pretrained ConvNet as foundation model
- remove final fully-connected layer (its outputs are the class scores from the pretraining task)
- pass training data (for the task at hand) through the network
- for each image, extract vector of last hidden layer's activations (e.g., 4096-dimensional for AlexNet)
- use the extracted values as features to train another model (to be adjusted to the task at hand)

(same idea as used before for SIFT features)

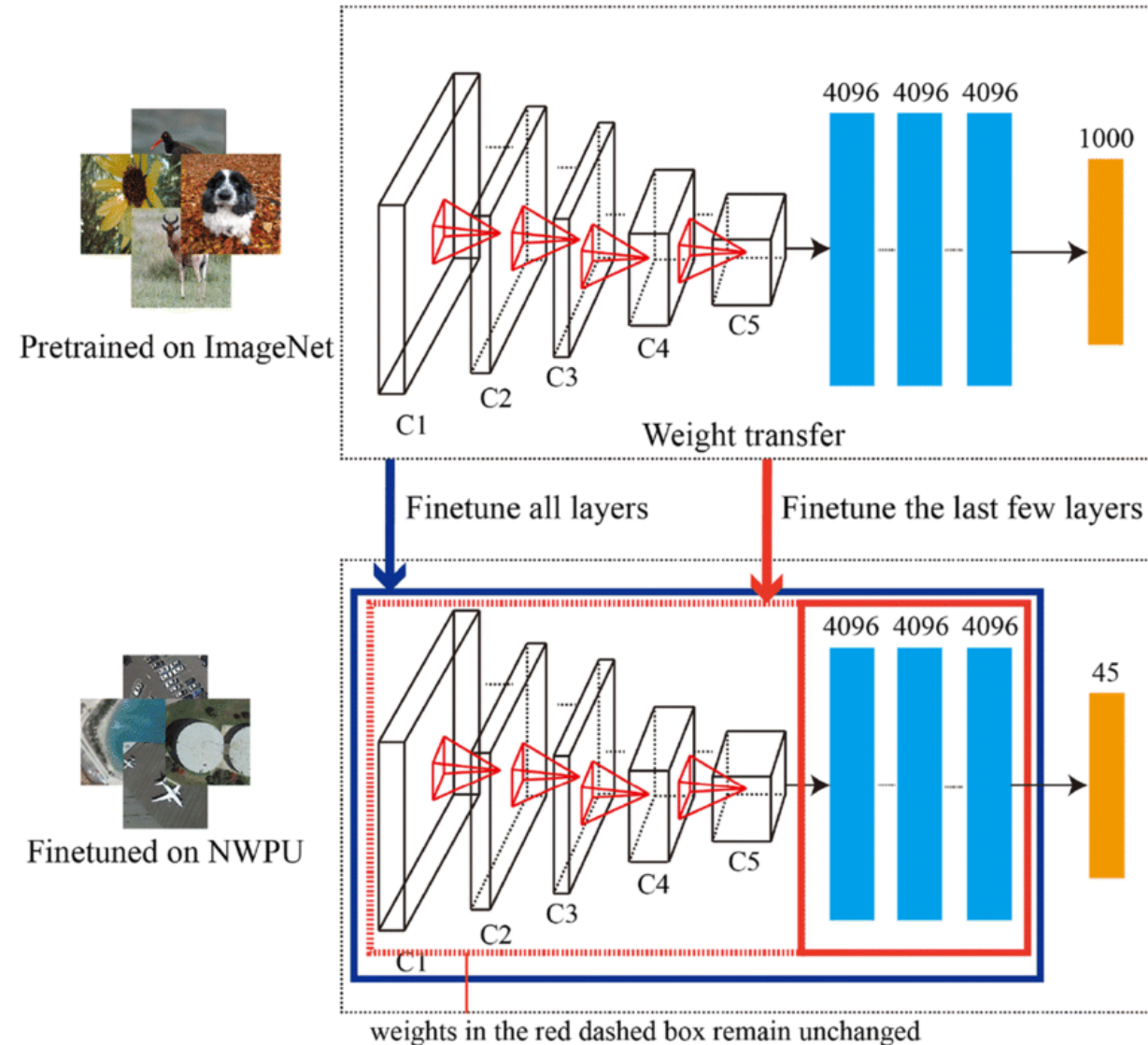
Finetuning

approach:

- get pretrained ConvNet as foundation model
- continue (starting with weights after pretraining) training with data for task at hand

typically, need to change architecture of final layer: different number of classes

options: finetune all weights or just some



Finetuning Scenarios

finetuning gives better adjustment to new task than feature extraction (important if new task differs a lot from pretraining task)

→ coming along with danger of overfitting

ConvNet features more generic in early layers (e.g., edges) and more specific to the data set of pretraining in later layers

→ for small finetuning data sets, typically keep weights of early layers fixed to avoid overfitting

for large finetuning data sets (less danger of overfitting), finetuning all layers can be beneficial

Recurrent Neural Networks (RNN)

speech recognition, natural language processing, time series, ...

problem: need to generalize across different points in time (or multiple positions of words within a sentence) and learn from context

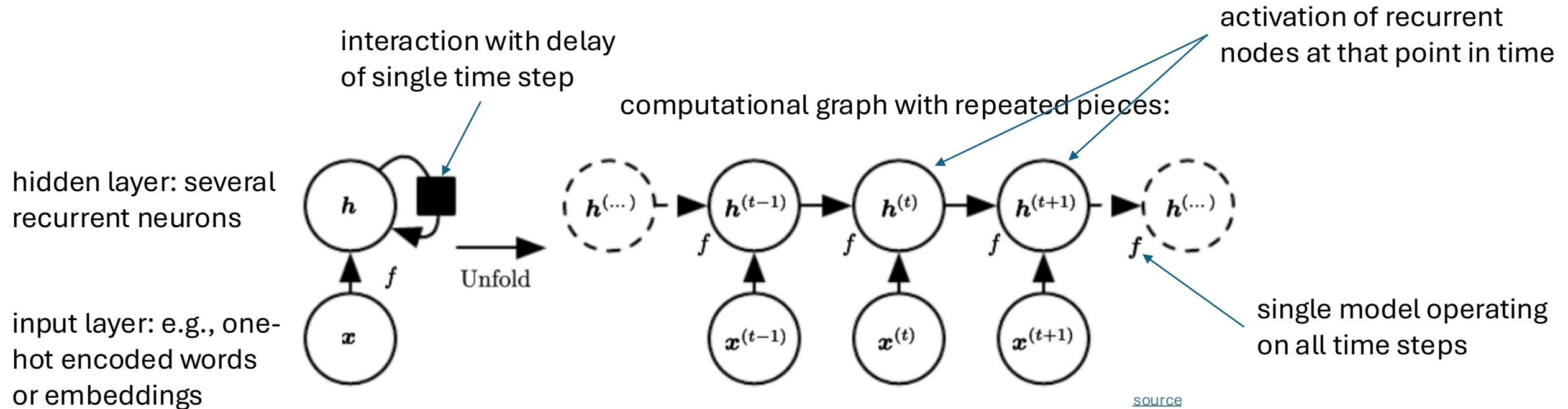
idea: parameter sharing across different parts of model

CNNs apply parameter sharing (convolutions) on grid-like structures (including 1-D grids like time series or speech), but this is limited to neighboring inputs (per layer).

→ need for approach to learn sequential structures (process input as stream, e.g., speech, rather than one batch, e.g., image)

Back-Propagation through Time

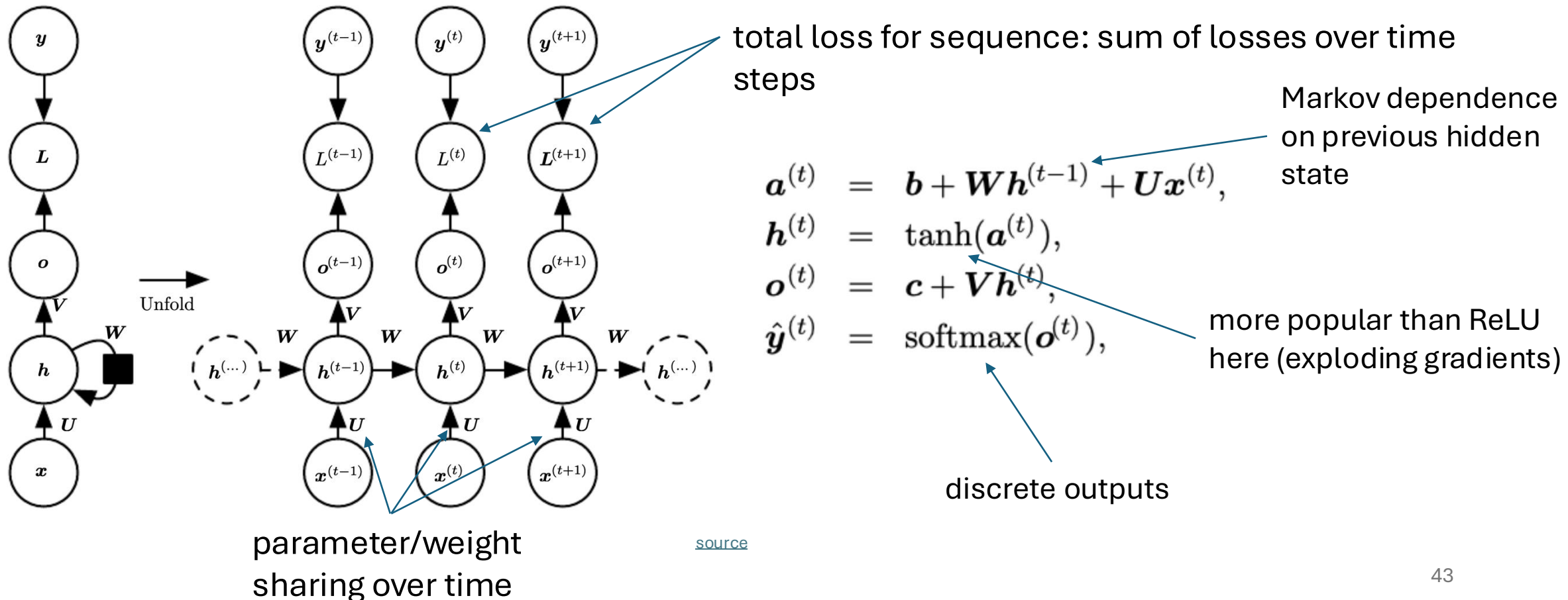
RNN: output of node directed back to input



different kind of depth: one recurrence for each sequential step (e.g., word)

Weight Sharing across Time

example: input-output mapping at each time step with recurrent hidden nodes



Further Examples

summarization of sequence:

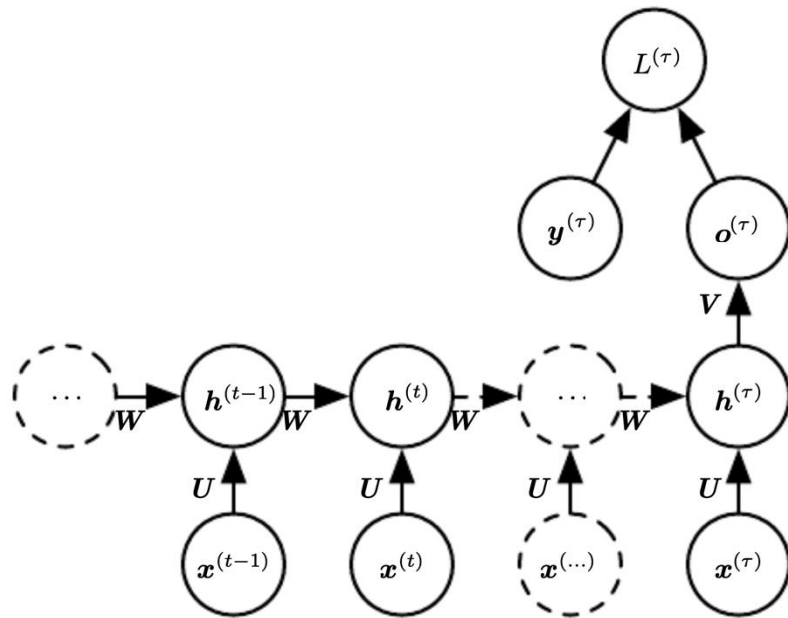
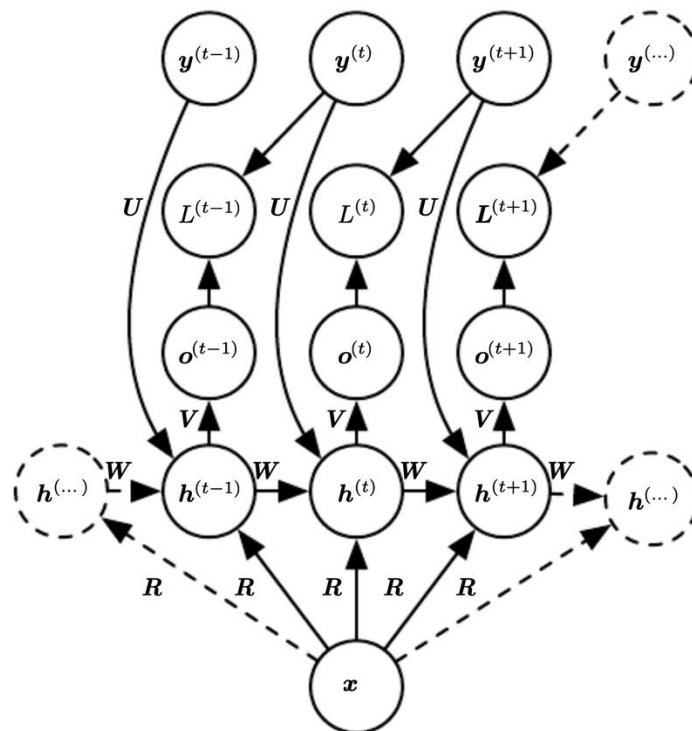
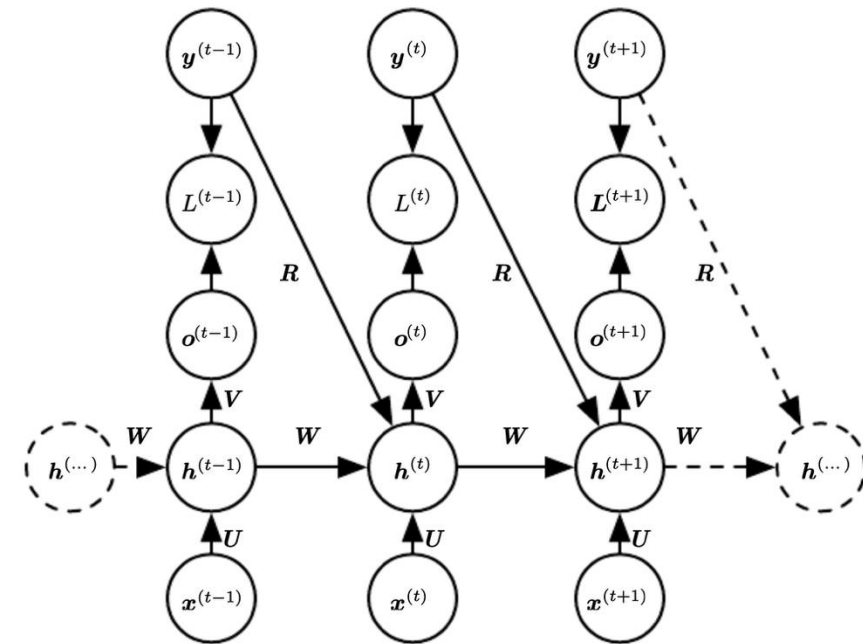


image captioning:



conditional sequence:



Visualization

neuron getting excited inside URLs (**excited**, not excited):

t	t	p	:	/	/	w	w	w	.	y	n	e	t	n	e	w	s	.	c	o	m	/	]	E	n	g	l	i	s	h	-	l	a	n	g	u	a	g	e		w	e	b	s	i	t	e		o	f		t
t	p	:	/	/	w	w	w	.	b	a	c	a	h	e	t	s	.	c	o	m	/	]	-	x	g	l	i	s	h		l	i	n	g	u	a	g	e	s	a	i	r	s	i	t	e		o	f		t	

next character prediction

high/low activation:

Cell sensitive to position in line:

The sole importance of the crossing of the Berezina lies in the fact that it plainly and indubitably proved the fallacy of all the plans for cutting off the enemy's retreat and the soundness of the only possible line of action--the one Kutuzov and the general mass of the army demanded--namely, simply to follow the enemy up. The French crowd fled at a continually increasing speed and all its energy was directed to reaching its goal. It fled like a wounded animal and it was impossible to block its path. This was shown not so much by the arrangements it made for crossing as by what took place at the bridges. When the bridges broke down, unarmed soldiers, people from Moscow and women with children who were with the French transport, all--carried on by vis inertiae--pressed forward into boats and into the ice-covered water and did not, surrender.

Cell that turns on inside quotes:

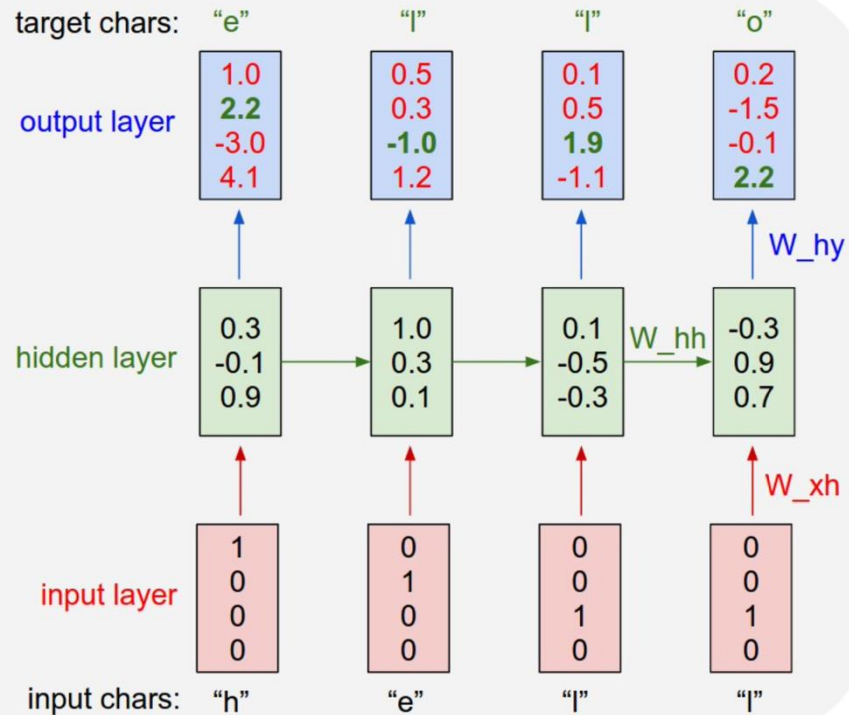
"You mean to imply that I have nothing to eat out of.... On the contrary, I can supply you with everything even if you want to give dinner parties," warmly replied Chichagov, who tried by every word he spoke to prove his own rectitude and therefore imagined Kutuzov to be animated by the same desire.

Kutuzov, shrugging his shoulders, replied with his subtle penetrating smile: "I meant merely to say what I said."

Cell that robustly activates inside if statements:

```
static int __dequeue_signal(struct sigpending *pending, sigset_t *mask,
                           siginfo_t *info)
{
    int sig = next_signal(pending, mask);
    if (sig) {
        if (current->notifier) {
            if (sigismember(current->notifier_mask, sig)) {
                if (!(current->notifier)(current->notifier_data)) {
                    clear_thread_flag(TIF_SIGPENDING);
                    return 0;
                }
            }
        }
        collect_signal(sig, pending, info);
    }
    return sig;
}
```

[source](#)



Gated RNNs

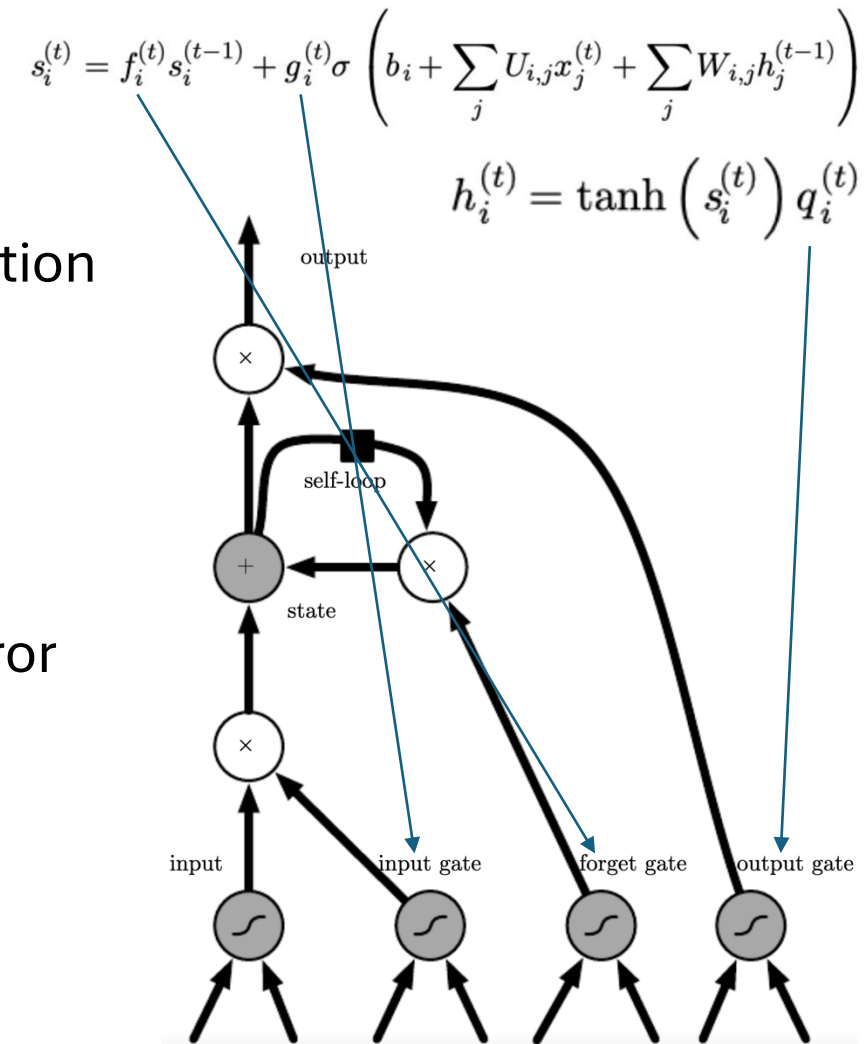
issue with long chain of gradients through recurrences:
vanishing (mainly) and exploding gradients in back-propagation
→ need to focus on important sequence elements

replace usual recurrent hidden nodes with long short-term
memory (LSTM) cells with internal recurrence (self-loop)

- linear self-loop in addition to outer recurrence of RNN: error carousel preserving (→ long) short-term memory (i.e., activation patterns)
- sigmoid activations with independent weights for input, forget, and output gates

other prominent gated RNN: gated recurrent unit (GRU)

long-term memory: weights
short term memory: activation patterns
(context)



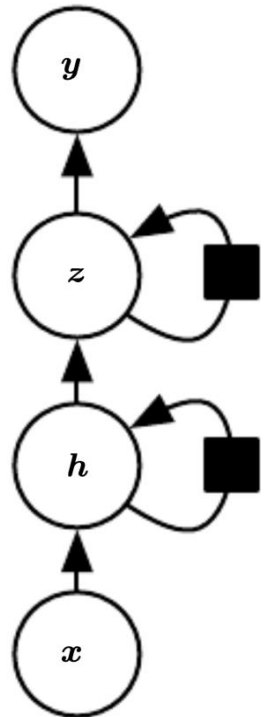
[source](#)

Issues with RNNs

- especially LSTMs very computationally expensive (with lots of parameters)
- transfer learning (using pre-trained network layers on new task, e.g., popular for CNNs) difficult

(hierarchical) representation learning: go deep by stacking several layers of recurrent nodes (e.g., important for [speech recognition](#)) → worsening efficiency even more

(self-)attention and transformers to the rescue ...



[source](#)

Tabular vs Unstructured Data

deep learning methods dominate applications on unstructured data (like text or images), but not necessarily on tabular data

typical characteristics of tabular data difficult to handle for deep learning:

- irregular patterns in target function (neural networks require piecewise continuous targets)
- uninformative features
- non-rotationally invariant data (linear combinations of features misrepresent the information)

tree-based models (e.g., gradient boosting) can naturally deal with these situations